

Sensation PM Report



Document #: SMS-08-00-002

Revision: B

Affected Products

SOMATOM Sensation 10/16/Cardiac/40/64/Cardiac64/Open

Reference Material

CT02-023.831.01.09.02 Sensation Maintenance Protocol

History

Date	Name	Rev	Change Descriptions
9/20/13	Document Control	A	Initial release
6/30/15	Document Control	B	Reformatted, removed unneeded sections

Approval

Department	Name	Date
Dept. Manager		
Engineering		
QA & Safety		
Document Control	Jhovany Hernandez	6/30/15

Preface

This document is valid proof of a Preventative Maintenance performed on the system. The Preventative Maintenance must be performed in the specified intervals, and the results are to be documented herein.

Refer to **CT02-023.831.01.09.02 Sensation Maintenance Protocol** for specific procedures.

This section addresses planned maintenance that should be performed by service personnel. Some of it is also listed in the Operator's Manual for the user to perform on daily and weekly basis.

However, service personnel should make sure that these procedures have been followed and if not, include them on the periodic visits.

Warning: All steps must be performed in accordance with the manufacturer's service manual.

This document must be completely filled out by the Field Service Engineer, i.e.:

- All boxes must be filled out. Use N/A where appropriate.
- Enter the Serial Number and the Date in the header of each page.
- After completing the check, fill out and sign the last page.

Serial #:

Date:

System Information

Customer:

Model:

System S/N:

System ID:

OEM System ID:

Measuring Equipment Used

Enter the measuring equipment used in the table below (meters, scopes, phantoms, etc.):

Equipment:

Serial #:

Serial #:

Date:

Safety Inspections

Function Tests

Pass	Fail	N/A	Action	Type
☐	☐	☐	Patient leakage current checked.	Safety Inspection
☐	☐	☐	Door contact switch checked.	Safety Inspection
☐	☐	☐	Radiation preparation indicators checked.	Safety Inspection
☐	☐	☐	Radiation indicators checked.	Safety Inspection
☐	☐	☐	Radiation switch-off checked. Measured value:	Safety Inspection
☐	☐	☐	Emergency STOP circuit checked.	Safety Inspection
☐	☐	☐	Tube cooling hoses checked.	Safety Inspection
☐	☐	☐	Safety limit switches checked.	Safety Inspection
☐	☐	☐	Function of spindle brake checked.	Safety Inspection
☐	☐	☐	Function of motor brake checked.	Safety Inspection
☐	☐	☐	Patient positioning aids checked.	Safety Inspection
☐	☐	☐	Circuit breaker checked.	Safety Inspection
☐	☐	☐	CT system condition checked.	Verification

Preventive Maintenance I

Patient Handling System

Pass	Fail	N/A	Action	Type
☐	☐	☐	Spindle nut lubricated.	Preventive Maintenance
☐	☐	☐	<i>Lower part of the scissor mechanism:</i> Linear guides lubricated.	Preventive Maintenance
☐	☐	☐	<i>Upper part of the scissor:</i> Linear guides lubricated.	Preventive Maintenance
☐	☐	☐	<i>Tabletop:</i> Guide rails lubricated.	Preventive Maintenance
☐	☐	☐	<i>Top support:</i> Guide rails lubricated.	Preventive Maintenance
☐	☐	☐	Movement force for horizontal tabletop travel checked.	Preventive Maintenance

Serial #:

Date:

Patient Handling System (Continued)

Pass	Fail	N/A	Action	Type
☐	☐	☐	Movement force for horizontal top support travel checked.	Preventive Maintenance
☐	☐	☐	Compression spring checked.	Preventive Maintenance
☐	☐	☐	Vertical motor brake cleaned.	Preventive Maintenance
☐	☐	☐	CT System has been thoroughly inspected and meets all specifications.	Verification

Preventive Maintenance II**Care Vision Option**

Pass	Fail	N/A	Action	Type
☐	☐	☐	Weight compensation of the support arm checked.	Safety Inspection

Gantry

Pass	Fail	N/A	Action	Type
☐	☐	☐	Function of E-box fans checked.	Safety Inspection
☐	☐	☐	Function of HV transformer fans checked.	Safety Inspection
☐	☐	☐	Function of C-box fan checked.	Safety Inspection
☐	☐	☐	Function of DMS fans checked.	Safety Inspection
☐	☐	☐	Slip ring compartment and assembly cleaned.	Preventive Maintenance
☐	☐	☐	Power brush block cleaned.	Preventive Maintenance
☐	☐	☐	Power carbon brushes checked.	Preventive Maintenance
☐	☐	☐	Power carbon brush replacement:	Preventive Maintenance
			Startup date:	
			Date of last replacement:	
☐	☐	☐	Power brush block replaced.	Preventive Maintenance
☐	☐	☐	Voltage and control data brushes checked.	Preventive Maintenance
☐	☐	☐	Voltage and/or control data brushes replaced.	Preventive Maintenance
			Startup date:	
			Date of last replacement:	

Serial #:

Date:

Gantry (Continued)

Pass	Fail	N/A	Action	Type
☐	☐	☐	Stator ring brush replaced.	Preventive Maintenance
			Startup date:	
			Date of last replacement:	
☐	☐	☐	Rack and pinion gear lubricated (only Sensation 64, Cardiac 64, not UHR-Plus).	Preventive Maintenance
☐	☐	☐	Detector window cleaned and checked.	Preventive Maintenance
			Startup date:	
			Date of last replacement:	
☐	☐	☐	Main bearing lubricated.	Preventive Maintenance
☐	☐	☐	Filter in dehumidifier replaced.	Preventive Maintenance
			Startup date:	
			Date of last replacement:	
☐	☐	☐	Maintenance of this section was performed completely.	Verification
			Name:	
			Date:	

Power Distribution Cabinet

Pass	Fail	N/A	Action	Type
☐	☐	☐	Air filter in PDC replaced.	Preventive Maintenance
			Startup date:	
			Date of last replacement:	
☐	☐	☐	Varistors/discharger was checked/replaced.	System Quality, Image Quality
☐	☐	☐	Maintenance of this section was performed completely.	Verification
			Name:	
			Date:	

Serial #:

Date:

Water Cooling System

Pass	Fail	N/A	Action		Type
☐	☐	☐	Water pressure checked. Measured value:		Preventive Maintenance
☐	☐	☐	Filter cleaned.		Preventive Maintenance
☐	☐	☐	Maintenance of this section was performed completely.		Verification
			Name:		
			Date:		

Imaging System

Pass	Fail	N/A	Action		Type
☐	☐	☐	Air inlets of ICS, and IES cleaned.		Preventive Maintenance
☐	☐	☐	Air filter in PDC replaced.		Preventive Maintenance
			Startup date:		
			Date of last replacement:		
☐	☐	☐	Maintenance of this section was performed completely.		Verification
			Name:		
			Date:		

System Testing Conclusion

Pass	Fail	N/A	Action	Type
☐	☐	☐	CT System has been thoroughly inspected and meets all specifications.	System Quality, Image Quality

Preventative Parts Replacement

Uninterrupted Power Supply

Pass	Fail	N/A	Action		Type
☐	☐	☐	UPS has been replaced.		Preventive Maintenance
			Startup date:		
			Date of last replacement:		
☐	☐	☐	Maintenance of this section was performed completely.		Verification
			Name:		
			Date:		

Serial #:

Date:

Image Quality

Pass	Fail	N/A	Action	Type
☐	☐	☐	A quality check was performed on the system.	System Quality, Image Quality
☐	☐	☐	Maintenance of this section was performed completely.	Verification
☐	☐	☐	Name:	
☐	☐	☐	Date:	

Final Steps

Unit Covers

Pass	Fail	N/A	Action	Type
☐	☐	☐	Covers and protective conductor cables are installed.	Mechanical Safety Inspection

System Protective Conductor - SOMATOM Sensation 10/16/Cardiac/40/64/Cardiac64/Open

Pass	Fail	N/A	Action	Type
☐	☐	☐	Gantry: Protective conductor resistance ≤ 200 m Ω . Measured value:	Electrical Safety Inspection
☐	☐	☐	Patient table (PHS): Protective conductor resistance ≤ 200 m Ω . Measured value:	Electrical Safety Inspection
☐	☐	☐	ICS tower: Protective conductor resistance ≤ 200 m Ω . Measured value:	Electrical Safety Inspection
☐	☐	☐	IRS cabinet: Protective conductor resistance ≤ 200 m Ω . Measured value:	Electrical Safety Inspection
☐	☐	☐	ICS monitor: Protective conductor resistance ≤ 200 m Ω . Measured value:	Electrical Safety Inspection
☐	☐	☐	IES monitor: Protective conductor resistance ≤ 200 m Ω . Measured value:	Electrical Safety Inspection
☐	☐	☐	PDC cabinet: Protective conductor resistance ≤ 200 m Ω . Measured value:	Electrical Safety Inspection
☐	☐	☐	WCS cabinet: Protective conductor resistance ≤ 200 m Ω . Measured value:	Electrical Safety Inspection

System Protective Conductor - Care Vision CT Option

Pass	Fail	N/A	Action	Type
☐	☐	☐	Overhead support: Protective conductor resistance ≤ 200 m Ω . Measured value:	Electrical Safety Inspection
☐	☐	☐	Monitor trolley: Protective conductor resistance ≤ 200 m Ω . Measured value:	Electrical Safety Inspection

Serial #:

Date:

Monitors

Pass	Fail	N/A	Action	Type
☐	☐	☐	All monitors cleaned, inspected for damage.	Mechanical Safety Inspection

System Test

Pass	Fail	N/A	Action	Type
☐	☐	☐	Condition of the CT system checked.	System Quality, Image Quality
☐	☐	☐	Maintenance of this section was performed completely.	Verification
			Name:	
			Date:	

Serial #:

Date:

System Condition

After performing scheduled PM, an evaluation of the system is performed with the following result (check one):

☐

No deficiencies. The image quality test resulted in no differences from required reference values.

☐

Slight deficiencies. There is no effect on operation; however they should be corrected preventively. The image quality test resulted in no differences from required reference values.

☐

Serious deficiencies. This must be corrected before operation of the system is permitted.

Field Service Engineer

Name:

Date:

Notes (Additional work to be performed, partial PM, customer requests):

Customer

Name:

Date: